Add capacity (disks) to a local tier (aggregate)

ONTAP 9

NetApp
March 27, 2024

This PDF was generated from https://docs.netapp.com/us-en/ontap/disks-aggregates/add-capacity-local-tier-overview-task.html on March 27, 2024. Always check docs.netapp.com for the latest.

Table of Contents

Add capacity (disks) to a local tier (aggregate) . . . 1
Add capacity (disks) to a local tier (aggregate) . . . 1
Workflow to add capacity to a local tier (expanding an aggregate) . . . 1
Methods to create space in an local tier (aggregate) . . . 2
Add capacity to a local tier (add disks to an aggregate) . . . 3
Add drives to a node or shelf . . . 10
Correct misaligned spare partitions . . . 12

Add capacity (disks) to a local tier (aggregate)

Add capacity (disks) to a local tier (aggregate)

Using different methods, you follow a specific workflow to add capacity.

- Workflow to add capacity to a local tier (aggregate)
- Methods to create space in a local tier (aggregate)

You can add disks to a local tier and add drives to a node or shelf.

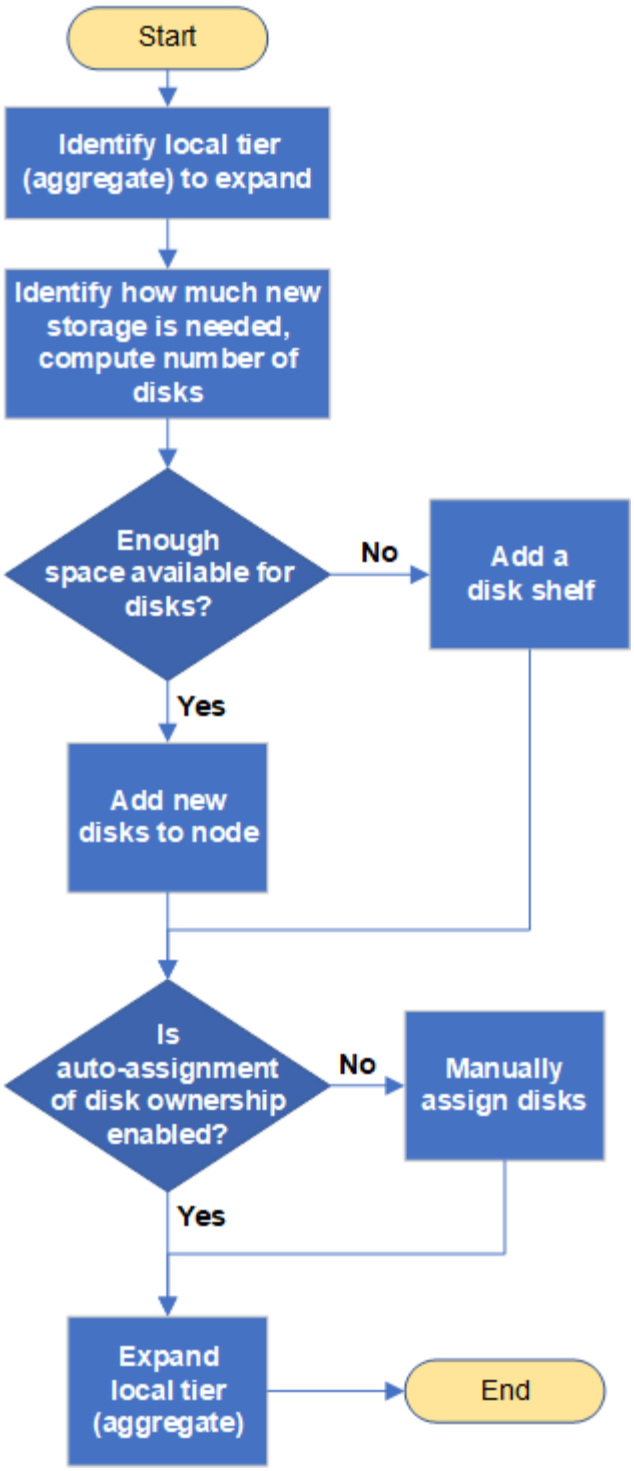
If needed, you can correct misaligned spare partitions.

- Add disks to a local tier (aggregate)
- Add drives to a node or shelf
- Correct misaligned spare partitions

Workflow to add capacity to a local tier (expanding an aggregate)

To add capacity to a local tier (expand an aggregate) you must first identify which local tier you want to add to, determine how much new storage is needed, install new disks, assign disk ownership, and create a new RAID group, if needed.

You can use either System Manager or the CLI to add capacity.



Methods to create space in an local tier (aggregate)

If a local tier (aggregate) runs out of free space, various problems can result that range from loss of data to disabling a volume's guarantee. There are multiple ways to make more space in a local tier.

All of the methods have various consequences. Prior to taking any action, you should read the relevant section in the documentation.

The following are some common ways to make space in local tier, in order of least to most consequences:

- Add disks to the local tier.
- Move some volumes to another local tier with available space.
- Shrink the size of volume-guaranteed volumes in the local tier.
- Delete unneeded volume Snapshot copies if the volume’s guarantee type is “none”.
- Delete unneeded volumes.
- Enable space-saving features, such as deduplication or compression.
- (Temporarily) disable features that are using a large amount of metadata .

Add capacity to a local tier (add disks to an aggregate)

You can add disks to an local tier (aggregate) so that it can provide more storage to its associated volumes.

System Manager (ONTAP 9.8 and later)

Use System Manager to add capacity (ONTAP 9.8 and later)

You can add capacity to a local tier by adding capacity disks.

> Beginning with ONTAP 9.12.1, you can use System Manager to view the committed capacity of a local tier to determine if additional capacity is required for the local tier. See Monitor capacity in System Manager.

About this task

You perform this task only if you have installed ONTAP 9.8 or later. If you installed an earlier version of ONTAP, refer to the tab (or section) labeled "System Manager (ONTAP 9.7 and earlier)
".

Steps

1. Click **Storage > Tiers**.
2. Click ⋮ next to the name of the local tier to which you want to add capacity.
3. Click **Add Capacity**.

   > If there are no spare disks that you can add, then the **Add Capacity** option is not shown, and you cannot increase the capacity of the local tier.

4. Perform the following steps, based on the version of ONTAP that is installed:

If this version of ONTAP is installed…	Perform these steps…
ONTAP 9.8, 9.9, or 9.10.1	1. If the node contains multiple storage tiers, then select the number of disks you want to add to the local tier. Otherwise, if the node contains only a single storage tier, the added capacity is estimated automatically. 2. Click **Add**.
Beginning with ONTAP 9.11.1	1. Select the disk type and number of disks. 2. If you want to add disks to a new RAID group, check the check box. The RAID allocation is displayed. 3. Click **Save**.

5. (Optional) The process takes some time to complete. If you want to run the process in the background, select **Run in Background**.
6. After the process completes, you can view the increased capacity amount in the local tier information at **Storage > Tiers**.

System Manager (ONTAP 9.7 and earlier)

Use System Manager to add capacity (ONTAP 9.7 and earlier)

You can add capacity to a local tier (aggregate) by adding capacity disks.

About this task

You perform this task only if you have installed ONTAP 9.7 or earlier. If you installed ONTAP 9.8 or later, refer to Use System Manager to add capacity (ONTAP 9.8 or later).

Steps

1. (For ONTAP 9.7 only) Click **(Return to classic version)**.
2. Click **Hardware and Diagnostics > Aggregates**.
3. Select the aggregate to which you want to add capacity disks, and then click **Actions > Add Capacity**.

   > You should add disks that are of the same size as the other disks in the aggregate.

4. (For ONTAP 9.7 only) Click **Switch to the new experience**.
5. Click **Storage > Tiers** to verify the size of the new aggregate.

CLI

Use the CLI to add capacity

The procedure for adding partitioned disks to an aggregate is similar to the procedure for adding unpartitioned disks.

What you'll need

You must know what the RAID group size is for the aggregate you are adding the storage to.

About this task

When you expand an aggregate, you should be aware of whether you are adding partition or unpartitioned disks to the aggregate. When you add unpartitioned drives to an existing aggregate, the size of the existing RAID groups is inherited by the new RAID group, which can affect the number of parity disks required. If an unpartitioned disk is added to a RAID group composed of partitioned disks, the new disk is partitioned, leaving an unused spare partition.

When you provision partitions, you must ensure that you do not leave the node without a drive with both partitions as spare. If you do, and the node experiences a controller disruption, valuable information about the problem (the core file) might not be available to provide to the technical support.

> Do not use the `disklist` command to expand your aggregates. This could cause partition misalignment.

Steps

1. Show the available spare storage on the system that owns the aggregate:

   ```
   storage aggregate show-spare-disks -original-owner node_name
   ```

   You can use the `-is-disk-shared` parameter to show only partitioned drives or only unpartitioned drives.

```
cl1-s2::> storage aggregate show-spare-disks -original-owner cl1-s2
-is-disk-shared true

Original Owner: cl1-s2
 Pool0
  Shared HDD Spares
                                                             Local
Local
                                                              Data
Root Physical
 Disk                        Type     RPM Checksum         Usable
Usable     Size Status
 --------------------------- ----- ------ -------------- --------
-------- -------- --------
 1.0.1                       BSAS    7200 block           753.8GB
73.89GB  828.0GB zeroed
 1.0.2                       BSAS    7200 block           753.8GB
0B  828.0GB zeroed
 1.0.3                       BSAS    7200 block           753.8GB
0B  828.0GB zeroed
 1.0.4                       BSAS    7200 block           753.8GB
0B  828.0GB zeroed
 1.0.8                       BSAS    7200 block           753.8GB
0B  828.0GB zeroed
 1.0.9                       BSAS    7200 block           753.8GB
0B  828.0GB zeroed
 1.0.10                      BSAS    7200 block                0B
73.89GB  828.0GB zeroed
2 entries were displayed.
```

2. Show the current RAID groups for the aggregate:

   `storage aggregate show-status` *`aggr_name`*

```
cl1-s2::> storage aggregate show-status -aggregate data_1

Owner Node: cl1-s2
 Aggregate: data_1 (online, raid_dp) (block checksums)
  Plex: /data_1/plex0 (online, normal, active, pool0)
   RAID Group /data_1/plex0/rg0 (normal, block checksums)
                                                   Usable Physical
     Position Disk        Pool Type     RPM     Size     Size Status
     -------- ----------- ---- ----- ------ -------- --------
----------
     shared   1.0.10        0   BSAS    7200  753.8GB  828.0GB
(normal)
     shared   1.0.5         0   BSAS    7200  753.8GB  828.0GB
(normal)
     shared   1.0.6         0   BSAS    7200  753.8GB  828.0GB
(normal)
     shared   1.0.11        0   BSAS    7200  753.8GB  828.0GB
(normal)
     shared   1.0.0         0   BSAS    7200  753.8GB  828.0GB
(normal)
5 entries were displayed.
```

3. Simulate adding the storage to the aggregate:

   `storage aggregate add-disks -aggregate` *`aggr_name`* `-diskcount` *`number_of_disks_or_partitions`* `-simulate true`

   You can see the result of the storage addition without actually provisioning any storage. If any warnings are displayed from the simulated command, you can adjust the command and repeat the simulation.

```
cl1-s2::> storage aggregate add-disks -aggregate aggr_test
-diskcount 5 -simulate true

Disks would be added to aggregate "aggr_test" on node "cl1-s2" in
the
following manner:

First Plex

  RAID Group rg0, 5 disks (block checksum, raid_dp)
                                                            Usable
Physical
    Position   Disk                      Type           Size
Size
    ---------- ------------------------- ---------- --------
--------
    shared     1.11.4                    SSD         415.8GB
415.8GB
    shared     1.11.18                   SSD         415.8GB
415.8GB
    shared     1.11.19                   SSD         415.8GB
415.8GB
    shared     1.11.20                   SSD         415.8GB
415.8GB
    shared     1.11.21                   SSD         415.8GB
415.8GB

Aggregate capacity available for volume use would be increased by
1.83TB.
```

4. Add the storage to the aggregate:

   `storage aggregate add-disks -aggregate` *`aggr_name`* `-raidgroup new -diskcount` *`number_of_disks_or_partitions`*

   When creating a Flash Pool aggregate, if you are adding disks with a different checksum than the aggregate, or if you are adding disks to a mixed checksum aggregate, you must use the `-checksumstyle` parameter.

   If you are adding disks to a Flash Pool aggregate, you must use the `-disktype` parameter to specify the disk type.

   You can use the `-disksize` parameter to specify a size of the disks to add. Only disks with approximately the specified size are selected for addition to the aggregate.

```
cl1-s2::> storage aggregate add-disks -aggregate data_1 -raidgroup
new -diskcount 5
```

5. Verify that the storage was added successfully:

   `storage aggregate show-status -aggregate` *`aggr_name`*

```
cl1-s2::> storage aggregate show-status -aggregate data_1

Owner Node: cl1-s2
 Aggregate: data_1 (online, raid_dp) (block checksums)
  Plex: /data_1/plex0 (online, normal, active, pool0)
   RAID Group /data_1/plex0/rg0 (normal, block checksums)
                                                                  Usable
Physical
     Position Disk                        Pool Type     RPM     Size
Size Status
     -------- --------------------------- ---- ----- ------ --------
-------- ----------
     shared   1.0.10                       0   BSAS    7200  753.8GB
828.0GB (normal)
     shared   1.0.5                        0   BSAS    7200  753.8GB
828.0GB (normal)
     shared   1.0.6                        0   BSAS    7200  753.8GB
828.0GB (normal)
     shared   1.0.11                       0   BSAS    7200  753.8GB
828.0GB (normal)
     shared   1.0.0                        0   BSAS    7200  753.8GB
828.0GB (normal)
     shared   1.0.2                        0   BSAS    7200  753.8GB
828.0GB (normal)
     shared   1.0.3                        0   BSAS    7200  753.8GB
828.0GB (normal)
     shared   1.0.4                        0   BSAS    7200  753.8GB
828.0GB (normal)
     shared   1.0.8                        0   BSAS    7200  753.8GB
828.0GB (normal)
     shared   1.0.9                        0   BSAS    7200  753.8GB
828.0GB (normal)
10 entries were displayed.
```

6. Verify that the node still has at least one drive with both the root partition and the data partition as spare:

   `storage aggregate show-spare-disks -original-owner` *`node_name`*

```
cl1-s2::> storage aggregate show-spare-disks -original-owner cl1-s2
-is-disk-shared true

Original Owner: cl1-s2
 Pool0
  Shared HDD Spares
                                                             Local
Local
                                                              Data
Root Physical
 Disk                        Type     RPM Checksum         Usable
Usable     Size Status
 --------------------------- ----- ------ -------------- --------
-------- -------- --------
 1.0.1                       BSAS    7200 block           753.8GB
73.89GB  828.0GB zeroed
 1.0.10                      BSAS    7200 block               0B
73.89GB  828.0GB zeroed
2 entries were displayed.
```

Add drives to a node or shelf

You add drives to a node or shelf to increase the number of hot spares or to add space to local tier (aggregate).

Before you begin

The drive you want to add must be supported by your platform. You can confirm using the NetApp Hardware Universe.

The minimum number of drives you should add in a single procedure is six. Adding a single drive might reduce performance.

Steps for the NetApp Hardware Universe

1. In the **Products** dropdown menu, select your hardware configuration
2. Select your platform.
3. Select the version of ONTAP you are running then **Show Results**.
4. Beneath the graphic, select **Click here to see alternate views**. Choose the view that matches your configuration.

Steps to install the drives

1. Check the NetApp Support Site for newer drive and shelf firmware and Disk Qualification Package files.

   If your node or shelf does not have the latest versions, update them before installing the new drive.

   Drive firmware is automatically updated (nondisruptively) on new drives that do not have current firmware versions.

2. Properly ground yourself.
3. Gently remove the bezel from the front of the platform.
4. Identify the correct slot for the new drive.

   > The correct slots for adding drives vary depending on the platform model and ONTAP version. In some cases you need to add drives to specific slots in sequence. For example, in an AFF A800 you add the drives at specific intervals leaving clusters of empty slots. Whereas, in an AFF A220 you add new drives to the next empty slots running from the outside towards the middle of the shelf.

   Refer to the steps in **Before you begin** to identify the correct slots for your configuration in the NetApp Hardware Universe.

5. Insert the new drive:
   a. With the cam handle in the open position, use both hands to insert the new drive.
   b. Push until the drive stops.
   c. Close the cam handle so that the drive is fully seated into the mid plane and the handle clicks into place. Be sure to close the cam handle slowly so that it aligns correctly with the face of the drive.
6. Verify that the drive's activity LED (green) is illuminated.

   When the drive's activity LED is solid, it means that the drive has power. When the drive's activity LED is blinking, it means that the drive has power and I/O is in progress. If the drive firmware is automatically updating, the LED blinks.

7. To add another drive, repeat Steps 4 through 6.

   The new drives are not recognized until they are assigned to a node. You can assign the new drives manually, or you can wait for ONTAP to automatically assign the new drives if your node follows the rules for drive auto-assignment.

8. After the new drives have all been recognized, verify that they have been added and their ownership is specified correctly.

Steps to confirm installation

1. Display the list of disks:

```
storage aggregate show-spare-disks
```

You should see the new drives, owned by the correct node.

2. **Optionally (for ONTAP 9.3 and earlier only),** zero the newly added drives:

```
storage disk zerospares
```

Drives that have been used previously in an ONTAP local tier (aggregate) must be zeroed before they can be added to another aggregate. In ONTAP 9.3 and earlier, zeroing can take hours to complete, depending on the size of the non-zeroed drives in the node. Zeroing the drives now can prevent delays in case you need to quickly increase the size of an local tier. This is not an issue in ONTAP 9.4 or later where drives are zeroed using *fast zeroing* which takes only seconds.

Results

The new drives are ready. You can add them to a local tier (aggregate), place them onto the list of hot spares, or add them when you create a new local tier.

Correct misaligned spare partitions

When you add partitioned disks to a local tier (aggregate), you must leave a disk with both the root and data partition available as a spare for every node. If you do not and your node experiences a disruption, ONTAP cannot dump the core to the spare data partition.

Before you begin

You must have both a spare data partition and a spare root partition on the same type of disk owned by the same node.

Steps

1. Using the CLI, display the spare partitions for the node:

```
storage aggregate show-spare-disks -original-owner node_name
```

Note which disk has a spare data partition (spare_data) and which disk has a spare root partition (spare_root). The spare partition will show a non-zero value under the `Local Data Usable` or `Local Root Usable` column.

2. Replace the disk with a spare data partition with the disk with the spare root partition:

```
storage disk replace -disk spare_data -replacement spare_root -action start
```

You can copy the data in either direction; however, copying the root partition takes less time to complete.

3. Monitor the progress of the disk replacement:

```
storage aggregate show-status -aggregate aggr_name
```

4. After the replacement operation is complete, display the spares again to confirm that you have a full spare disk:

```
storage aggregate show-spare-disks -original-owner node_name
```

You should see a spare disk with usable space under both “Local Data Usable” and `Local Root Usable`.

Example

You display your spare partitions for node c1-01 and see that your spare partitions are not aligned:

```
c1::> storage aggregate show-spare-disks -original-owner c1-01

Original Owner: c1-01
 Pool0
  Shared HDD Spares
                              Local    Local
                               Data     Root  Physical
 Disk    Type   RPM Checksum Usable   Usable      Size
 ------- ----- ---- -------- ------- ------- --------
 1.0.1   BSAS  7200 block    753.8GB     0B   828.0GB
 1.0.10  BSAS  7200 block         0B 73.89GB  828.0GB
```

You start the disk replacement job:

```
c1::> storage disk replace -disk 1.0.1 -replacement 1.0.10 -action start
```

While you are waiting for the replacement operation to finish, you display the progress of the operation:

```
c1::> storage aggregate show-status -aggregate aggr0_1

Owner Node: c1-01
 Aggregate: aggr0_1 (online, raid_dp) (block checksums)
  Plex: /aggr0_1/plex0 (online, normal, active, pool0)
   RAID Group /aggr0_1/plex0/rg0 (normal, block checksums)
                                      Usable Physical
 Position Disk    Pool Type   RPM     Size     Size Status
 -------- ------- ---- ---- ----- -------- -------- ----------
 shared   1.0.1    0   BSAS  7200  73.89GB  828.0GB (replacing,copy in
progress)
 shared   1.0.10   0   BSAS  7200  73.89GB  828.0GB (copy 63% completed)
 shared   1.0.0    0   BSAS  7200  73.89GB  828.0GB (normal)
 shared   1.0.11   0   BSAS  7200  73.89GB  828.0GB (normal)
 shared   1.0.6    0   BSAS  7200  73.89GB  828.0GB (normal)
 shared   1.0.5    0   BSAS  7200  73.89GB  828.0GB (normal)
```

After the replacement operation is complete, confirm that you have a full spare disk:

```
ie2220::> storage aggregate show-spare-disks -original-owner c1-01

Original Owner: c1-01
 Pool0
  Shared HDD Spares
                              Local    Local
                               Data     Root  Physical
 Disk   Type   RPM Checksum Usable   Usable      Size
 ------ ----- ---- -------- -------- ------- --------
 1.0.1  BSAS  7200 block    753.8GB  73.89GB  828.0GB
```

Copyright information

Copyright © 2024 NetApp, Inc. All Rights Reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means—graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system—without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

LIMITED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (b)(3) of the Rights in Technical Data -Noncommercial Items at DFARS 252.227-7013 (FEB 2014) and FAR 52.227-19 (DEC 2007).

Data contained herein pertains to a commercial product and/or commercial service (as defined in FAR 2.101) and is proprietary to NetApp, Inc. All NetApp technical data and computer software provided under this Agreement is commercial in nature and developed solely at private expense. The U.S. Government has a non-exclusive, non-transferrable, nonsublicensable, worldwide, limited irrevocable license to use the Data only in connection with and in support of the U.S. Government contract under which the Data was delivered. Except as provided herein, the Data may not be used, disclosed, reproduced, modified, performed, or displayed without the prior written approval of NetApp, Inc. United States Government license rights for the Department of Defense are limited to those rights identified in DFARS clause 252.227-7015(b) (FEB 2014).

Trademark information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.